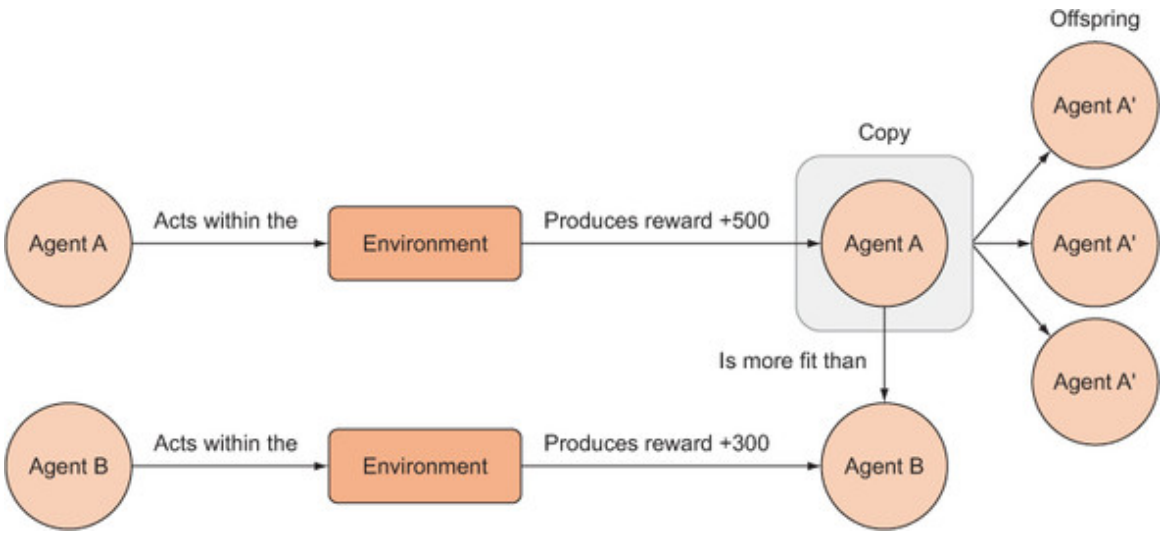
agent to be more similar to agent A than B. Remember, the only reason why agent A would be more fit than agent B is because its model parameters were slightly more optimized to the environment.

The objective in evolutionary reinforcement learning is exactly the same as in backpropagation and gradient descent-based training. The only difference is that we use this evolutionary process, which is often referred to as a *genetic algorithm*, to optimize the parameters of a model such as a neural network (figure 6.3).

Figure 6.3. In an evolutionary algorithm approach to reinforcement learning, agents compete in an environment, and the agents that are more fit (those that generate more rewards) are preferentially copied to produce offspring. After many iterations of this process, only the most fit agents are left.



The process is quite simple, but let's run through the steps of a genetic algorithm in more detail. Let's say we have a neural network that we want to use as an agent to play Gridworld, and we want to train it using a genetic algorithm. Remember, *training* a neural network just means iteratively updating its parameters such that its performance improves. Also recall that given a fixed neural network architecture, the parameters completely determine its behavior, so to copy a neural network we just need to copy its parameters.

Here's how we would train such a neural network using a genetic algorithm (graphically depicted in figure 6.4):